

Anna Shelegina

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Research Interests

Cartographic visual analytics and web-based geospatial interfaces for environmental decision-making, with a particular focus on agroforestry, sustainable land use, and soil information systems. Interested in bridging rigorous soil science with operationally meaningful spatial data products and decision-support tools.

Education

Erasmus Mundus Joint Master in Cartography

2024 – present

TU Munich · TU Wien · TU Dresden · TU Munich | joint degree, 4 semesters

Relevant coursework: Geo-information, Image Analysis for Mapping, Principles of Spatial Data Mining and Machine Learning, Geovisual Analytics, Remote Sensing based Environmental Mapping, Basics of Virtual Reality, LBS, Georelief and Cartography Morphogenetic and Environmental Understanding.

M.Sc. in Ecology and Land Management

2021 – 2023

Lomonosov Moscow State University (MSU) | GPA: 5.0 / 5.0

Thesis: Analysis of the characteristics of atmospheric microparticles of various genesis reflecting ecological and geographical conditions of their formation.

B.Sc. in Soil Science (Soil Geography Track)

2017 – 2021

Lomonosov Moscow State University (MSU) | GPA: 4.5 / 5.0

Thesis: Mineralogical composition of soils on various soil-forming rocks and solid atmospheric precipitation in the Moscow region.

Academic Experience

Visiting Researcher

Apr 2026 – present

Center for Sensing Solutions, Eurac Research | Bolzano, Italy

- Developing a cartographic visual analytics framework for agroforestry and WEFE-Nexus intervention planning as part of the EU Horizon Europe project **TRANS-SAHARA**.
- Implementing a multi-criteria suitability model (MCDA / Weighted Linear Combination) using open environmental datasets to support spatial land management decisions.
- Building a web-based map interface for suitability visualisation and intervention scenarios, grounded in cartographic theory and user-centred design principles.
- Integrating soil, climate, and land-use datasets to produce actionable spatial information products for policymakers and field practitioners.

Geospatial Analyst Intern, GIS Research Group

Oct 2023 – Jan 2024

Scientific Center “Smart Technologies for Sustainable Development of the Urban Environment”, RUDN University | Moscow, Russia

- Performed GIS- and remote-sensing-based spatial analysis of northern Russian territories, supporting environmental research on ecosystem dynamics under climate change.
- Designed thematic and analytical maps integrating anthropogenic pressures, natural conditions, and accessibility factors to inform spatial planning scenarios.

Research Assistant, RFBR Grant Project

Sept 2020 – May 2021

“Environmental Risks Caused by Atmospheric Microparticles”, Russian Foundation for Basic Research | Lomonosov Moscow State University

- Analyzed mineralogical composition of atmospheric microparticles using X-ray microscopy with spectrum analysis.
- Identified mineral types and structural characteristics of quartz and related particulates.

- Performed spatial zoning of the Russian Federation based on sediment types and mineralogical properties.
- Contributed to the classification of territories into glacial, aeolian, eluvial, and subaqueous deposition regions.

Undergraduate Research Trainee

2019 – 2020

Department of Soil Science, Lomonosov Moscow State University | Moscow, Russia

- Contributed to the Informational System “Soil-Geographic Database of the Russian Federation” – a national web-based cartographic platform for soil data.
- Digitized soil polygons from archival cartographic sources and integrated field survey data into the spatial database.

Professional Experience

Web Cartographer

July 2025 – Sept 2025

Sobrano v Sadu – major Russian apple producer | Remote

- Built an interactive web map that streamlined logistics across orchard sites and improved field-team decision-making.
- Implemented cadastral monitoring of orchard parcels by integrating land registry data with infrastructure, road networks, and field-level attributes.

GIS Specialist

Nov 2023 – Jul 2024

Geoscan Group | Moscow, Russia

- Processed orthophoto imagery and aerial survey data in Agisoft Metashape, generating dense point clouds, DTMs, and DSMs through photogrammetric workflows.
- Classified point clouds and digitized terrain features to produce large-scale topographic plans in accordance with cartographic and methodological standards.
- Interpreted Earth observation data and contributed to spatial analysis tasks supporting environmental monitoring projects.

Web Cartographer

Oct 2023 – Jan 2024

ATIS South Caucasus | Tbilisi, Georgia

- Built a web map visualizing the spatial structure of almond orchards in Georgia, integrating agrochemical research results and site inventory data.
- Designed cartographic workflows that translated field data into interpretable spatial insights to support orchard management decisions.

Publications

Peer-Reviewed Article

Kolesnikova, V.M., Salimgareeva, O.A., Ladonin, D.V., Vertyankina, V.Y., **Shelegina, A. S.** (2023). Morphological and mineralogical characteristics of atmospheric microparticles and chemical pollution of street dust in the Moscow Region. *Atmosphere*, 14(2), 403. <https://doi.org/10.3390/atmos14020403>

Cartographic Publications

Shelegina, A. S. (2025). Map contribution in: *Atlas of Sustainability*. International Cartographic Association (ICA). Open-access publication. icaci.org

Conference Papers

- Salimgareeva, O. A., Kolesnikova, V.M., **Shelegina, A. S.** (2023). *Features of natural and anthropogenic atmospheric microparticles and their ecological significance*. In *Man and Nature: Structural and Regional Problems of Socionatural Evolution* (Series: Socionatural History, Vol. 45, pp. 109–116). Institute of Geography, Russian Academy of Sciences.
- Kolesnikova, V.M., Salimgareeva, O. A., **Shelegina, A. S.** (2021). *Complex mineralogical and micromorphological analysis of atmospheric microparticles in an urban park environment (Istra, Russia)*. In *Ecology of River Basins: Proceedings of the 10th International Scientific and Practical Conference* (pp. 194–199). Vladimir State University named after A. G. and N. G. Stoletovs.

Teaching Experience

Teaching Assistant, Soil Chemistry Laboratory

Oct 2021 – Feb 2022

- Supervised undergraduate laboratory sessions in soil chemistry, guiding students through analytical procedures and data interpretation.
- Assisted in preparing experimental setups, instructing on laboratory techniques, and evaluating student performance.

Honours, Awards & Grants

Erasmus Mundus Joint Master Degree Scholarship

2024 – 2026

European Commission | awarded for the Cartography M.Sc. programme

Prize-Winner, Universiade in Soil Science and Ecology

2021

Lomonosov Moscow State University

Professional Memberships

Member, Soil Intelligence Working Group

2026 – present

International Soil Modeling Consortium (ISMC)

International working group developing data-driven approaches and AI methods for soil modeling and mapping.

Skills

Programming: Python (geospatial analysis, raster processing, image classification, MCDA workflows), R (spatial statistics), JavaScript (web mapping, React), HTML/CSS, SQL

GIS & Remote Sensing: QGIS, ArcGIS Pro, ArcGIS Online, Google Earth Engine, PostGIS, Agisoft Metashape (photogrammetry, point cloud classification, DTM/DSM)

Web Mapping: Leaflet, Mapbox GL JS, MapLibre, Turf.js

Cartographic & Visual Design: Adobe Illustrator, Adobe Photoshop, Blender, Figma; thematic mapping, visual hierarchy, user-centred interface design

Languages

English (C1) · German (A2) · Spanish (A1) · Russian (Native)

References

Dr. Ekaterina Churikova

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